

Multiplying $(\sin x)(2\sin 2x)$

First we use the $\sin 2x$ identity:

$$\sin 2x = 2\sin x \cos x$$

$$\Rightarrow (\sin x)(2)(2\sin x \cos x)$$

$$\Rightarrow 2\sin x (2\sin x \cos x)$$

$$\Rightarrow \boxed{4\sin^2 x \cos x}$$

